Network Traffic Pattern Mining and Intrusion Classification

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Objective and project summary

Study prepared network-flow records through data quality checks, descriptive analysis, frequent-pattern mining and binary intrusion classification. Five miners are compared on identical transactions; a majority baseline and four learned classifiers are evaluated with training-only preprocessing and a held-out test partition.

The saved notebook completed all 17 code cells and its consistency checks in **189.3 seconds** on 10 October 2026. Random Forest was chosen using validation macro F1. Its test accuracy was **87.12%**, macro F1 **0.8649** and ROC-AUC **0.9814**. The report uses that execution and all seven original embedded figures.

Dataset origin and actual files

UNSW-NB15 was produced in the UNSW Canberra cyber range with IXIA PerfectStorm traffic generation and captured network traffic. This project uses the prepared training/testing partition documented by UNSW, downloaded from the Mireu-Lab Hugging Face mirror. [1, 2]

Local file in data/	Mirror file	Role / raw rows
UNSW_NB15_training-set.csv	test.csv	Training / 175,341
UNSW_NB15_testing-set.csv	train.csv	Testing / 82,332

The mirror filenames were reversed relative to the expected training/testing roles; the notebook already used the correct roles. The local names now make that mapping explicit. Downloads are pinned to revision **3eb02c4a0a29866b7abcb5ef77c45cf4fcc8f6b0**. Missing files go directly to their canonical destinations; cached files must match the expected hash, row count and schema.

Training SHA-256:
bec7dd5ec88dc2a0ccc7a07879d338395ed7421750f675fd0339e07dfe0648fa
Testing SHA-256:
734fe6642edf758f7c94d7d9149426b49d202fe8e7bf0bef47392489c3c0a559

Schema, population and provenance boundary

The CSVs have 45 columns: 39 numeric predictors, three categorical predictors (**proto, service, state**), and **id, attack_cat, label**. The target is label: 0 = Normal, 1 = Attack. The identifier and attack category are excluded from predictors. Byte counts, durations, rates, TTLs, timing and connection counts describe traffic; attack_cat is used only for interpretation.

Raw training contains 56,000 Normal and 119,341 Attack rows; testing contains 37,000 Normal and 45,332 Attack rows. The existing cleaning policy leaves **99,607 eligible training rows** and retains **82,332 test rows**.

An independent source review recorded in the notebook found exact agreement with the standard configuration of lacg030175/UNSW-NB15 for every row and all 44 common columns after excluding id. This verifies agreement between the two mirrors; direct identity with the original UNSW-hosted files remains unverified. Cleaning changes the training population, so published performance comparisons must account for that preprocessing difference. [3]

Data quality and experiment design

Existing cleaning policy

Categorical values are stripped; blank strings, ?, null and None become missing. The service token "-" remains a valid category. Numeric parsing failures, infinities and negative traffic values become missing. TTL values must be in 0-255 and is_sm_ips_ports must be binary. The observed is_ftp_login value 2 is retained as a feature code. Invalid targets are removed; imputation rules are learned later inside each model pipeline.

Both raw files had zero missing cells, parse failures, infinite values, negative values, domain violations and label/category disagreements. Thus the population change comes from identity handling, rather than bad-value removal. Identity uses all 42 predictors, excluding id and both label fields. Hash-based identities were checked against exact row deduplication; no collision was detected.

Training/test identity audit	Saved count
ambiguous training rows removed	940
ambiguous training identities	229
extra training predictor copies	74,301
training rows overlapping test (before other exclusions)	10,279
distinct identities common to raw cleaned splits	1,302
training rows removed by union of all policies	75,734
eligible training rows	99,607
valid test rows retained	82,332
unique test predictor identities	53,946

All training identities with conflicting labels are excluded, only one copy of a remaining training identity is kept, and identities appearing in test are removed from training. These exclusions overlap: their counts cannot be added. Test multiplicity is preserved. Raw exact duplicates excluding id were 67,601 training / 26,387 testing; extra predictor copies were 74,301 / 28,386.

Split and protection against target leakage

A stratified 80/20 split (seed 42) makes 79,685 fit rows and 19,922 validation rows. Test has 82,332 rows. Test predictors are consulted only for the specified identity-overlap audit; no test labels enter training or selection. The selector, imputation, category vocabulary, scaling and numeric cutpoints are fitted on training data. The primary transaction attributes exclude id, label and attack_cat.

A fit-only selector removes constants and near-redundant numeric columns using absolute Spearman correlation >= 0.995 on at most 20,000 seeded fit rows. No constants were found; ct_ftp_cmd and ct_flw_http_mthd were removed: 42 potential predictors became 40. After validation selection, pipelines are refitted on all 99,607 eligible training rows.

Primary input	Transformation actually used
Gini / entropy trees and forest	Numeric median imputation; categorical one-hot encoding (minimum frequency 20, unseen categories handled); float32 sparse representation.
Mixed Naive Bayes	Numeric median imputation, log1p and z-score scaling; categorical missing token and ordinal encoding with a reserved unseen code.
Pattern mining	Five numeric attributes become training-derived quantile intervals; three categorical attributes become feature=value tokens.
Separate course demonstrations	Min-max/decimal alternatives and rank-bin smoothing do not enter classifiers or transactions. IQR flags do not remove valid extremes.

Descriptive analysis of eligible training flows

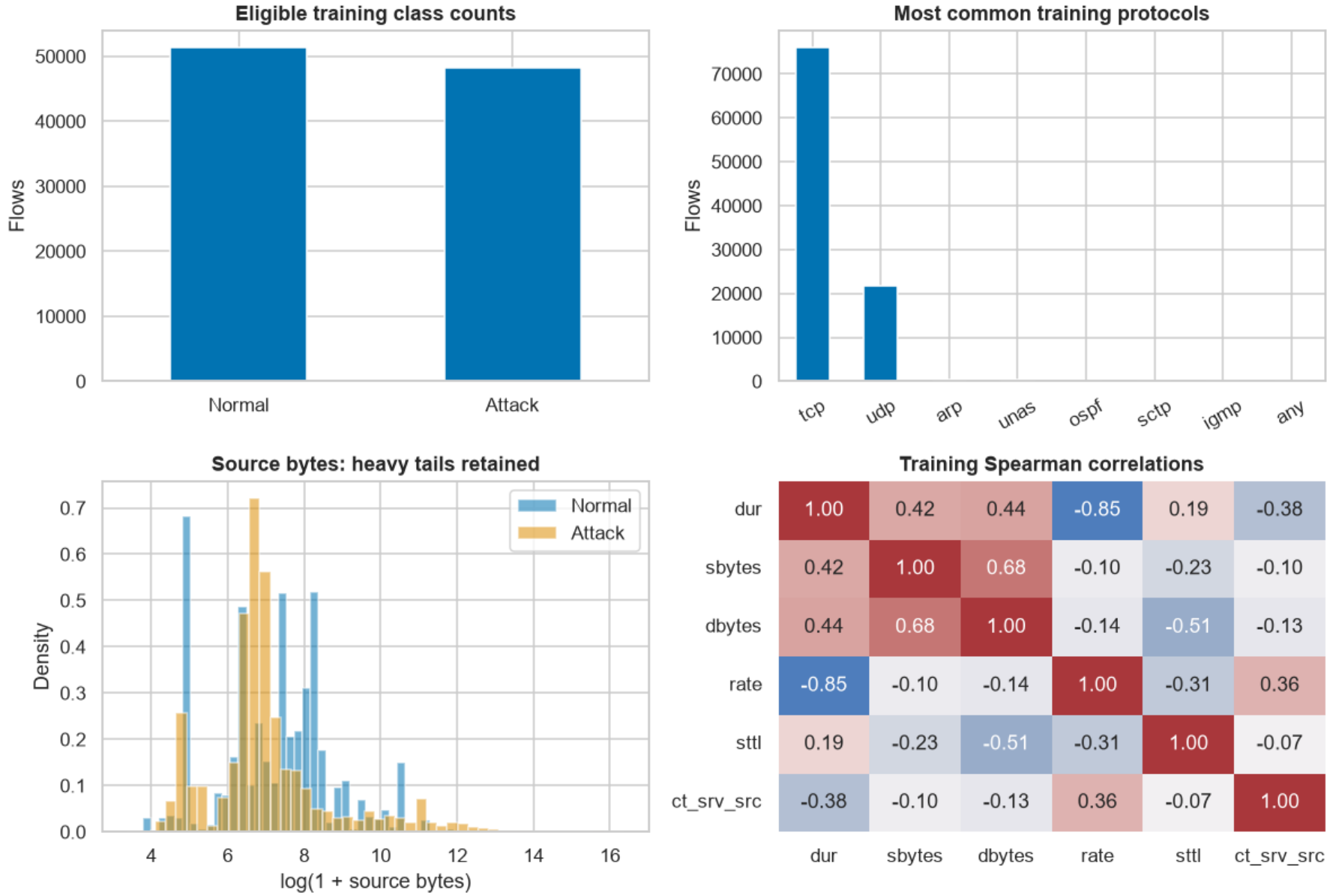


Figure 1. *All 99,607 eligible training flows: class and protocol counts, log-scaled source bytes and a Spearman correlation view. Attack prevalence is 48.4% after identity cleaning. TCP dominates; long byte tails and correlated traffic attributes support fit-only reduction and careful model preprocessing.*

Traffic distributions and class overlap

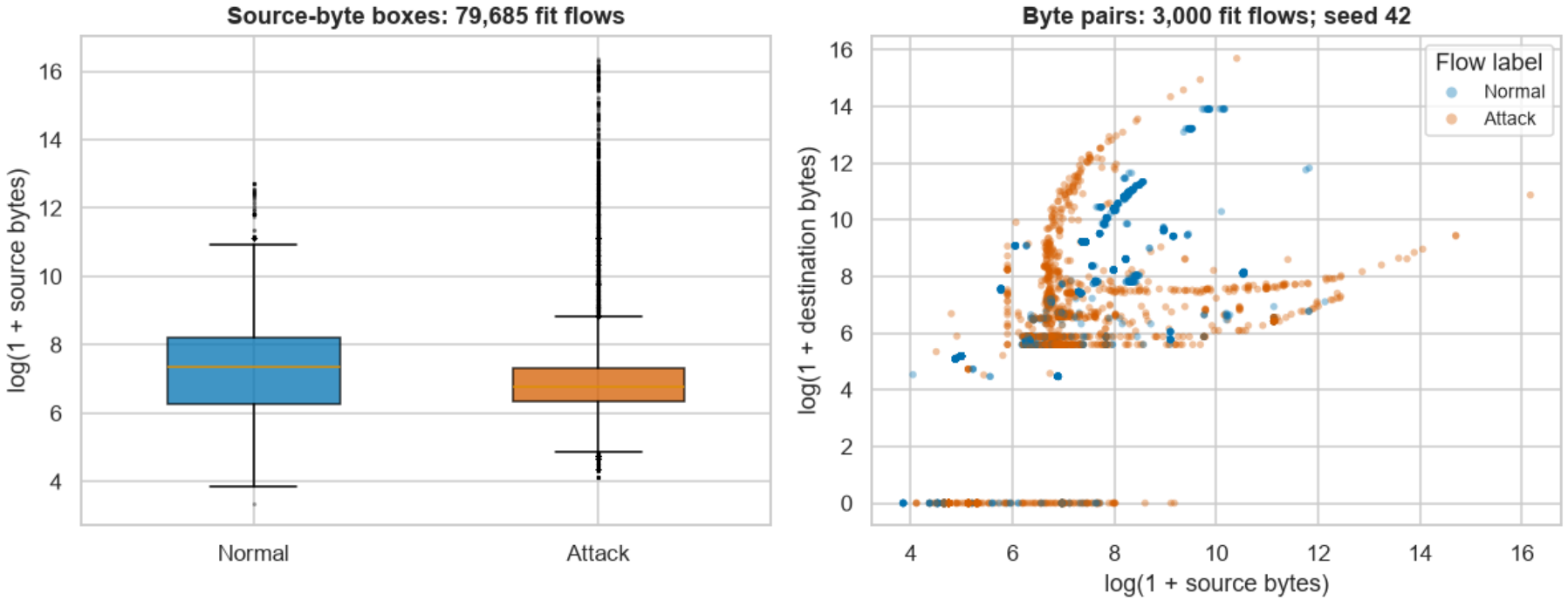


Figure 2. *The boxplot uses all 79,685 fit rows. The scatter uses 3,000 complete fit pairs sampled with seed 42. log1p is used for visual scale; observations are not clipped. The class distributions overlap, so a single byte threshold cannot perfectly separate normal and attack traffic.*

Eligible training attribute	Mean	Median	95th percentile	99th percentile
dur	1.277	0.388	3.316	26.881
sbytes	14785.029	1004.000	34710.000	136021.000
dbytes	25427.486	646.000	59460.000	645754.000
rate	26854.051	74.650	166666.661	500000.001

Source-byte log medians in the fit set are **7.340** (Normal, n=41092) and **6.772** (Attack, n=38593). The sampled log-byte correlations are 0.649 and 0.492. These are descriptive associations, not causal or class-separation guarantees.

Source-byte mean 14,785.029 exceeds its median 1,004.000; destination-byte mean 25,427.486 exceeds its median 646.000. Large valid transfers pull the means upward. Correlation also means that naive conditional-independence assumptions and individual importance rankings require caution.

Normalization and outlier diagnostics

Same-attribute normalization demonstration

All three alternatives use raw sbytes from the 79,685 fit rows. Min-max is (x-min)/(max-min); z-score is (x-mean)/population standard deviation (ddof=0); decimal scaling is $x/10^j$, with the smallest nonnegative integer j that makes every absolute value strictly below 1. Constants map to zero for min-max/z-score; all-zero decimal input uses j=0.

Rank example	Raw bytes	Min-max	Z-score	Decimal
minimum	28.0	0.000000	-0.063015	2.800000e-07
25% rank	544.0	0.000040	-0.060838	5.440000e-06
50% rank	1004.0	0.000075	-0.058897	1.004000e-05
75% rank	2934.0	0.000224	-0.050754	2.934000e-05
95% rank	34739.0	0.002677	0.083448	3.473900e-04
maximum	12965233.0	1.000000	54.643693	1.296523e-01

Representation	Mean	Population std.
original	14962.335345	236994.793381
min_max	0.001152	0.018279
z_score	-0.000000	1.000000
decimal	0.000150	0.002370

The largest raw value is 12,965,233 and decimal j=8. Its z-score is 54.643693: scaling does not erase the long tail. The notebook checks bounds, ordering, constant inputs (including extreme constants), zeros, negative constants and exact powers of ten. This comparison is a demonstration; only the NB pipeline uses its learned log1p/z-score sequence.

IQR fences and observed skewness

For each fit attribute, values strictly outside Q1-1.5 IQR and Q3+1.5 IQR are flagged. These flags describe tails and do not prove an error. All valid extremes remain in the primary data. Skewness is the adjusted Fisher-Pearson coefficient; raw and log1p values use the same rows.

Feature	IQR	Lower fence	Upper fence	Flagged	Flagged %
dur	0.9987	-1.4919	2.5030	5687	7.1369
sbytes	2390.0000	-3041.0000	6519.0000	9508	11.9320
dbytes	3112.0000	-4400.0000	8048.0000	16153	20.2711
rate	2909.7804	-4339.0302	7300.0914	10957	13.7504

Feature	Raw skewness	log1p skewness
dur	9.0503	2.6467
sbytes	34.5235	0.8940
dbytes	31.2261	-0.5205
rate	6.4640	0.7676

Absolute skewness fell for all four attributes in this execution. Destination-byte log skewness became negative (-0.5205), so the transform can change tail shape rather than simply make data symmetric. This observed reduction is not a promise for every variable or dataset.

Equal-frequency rank-bin smoothing

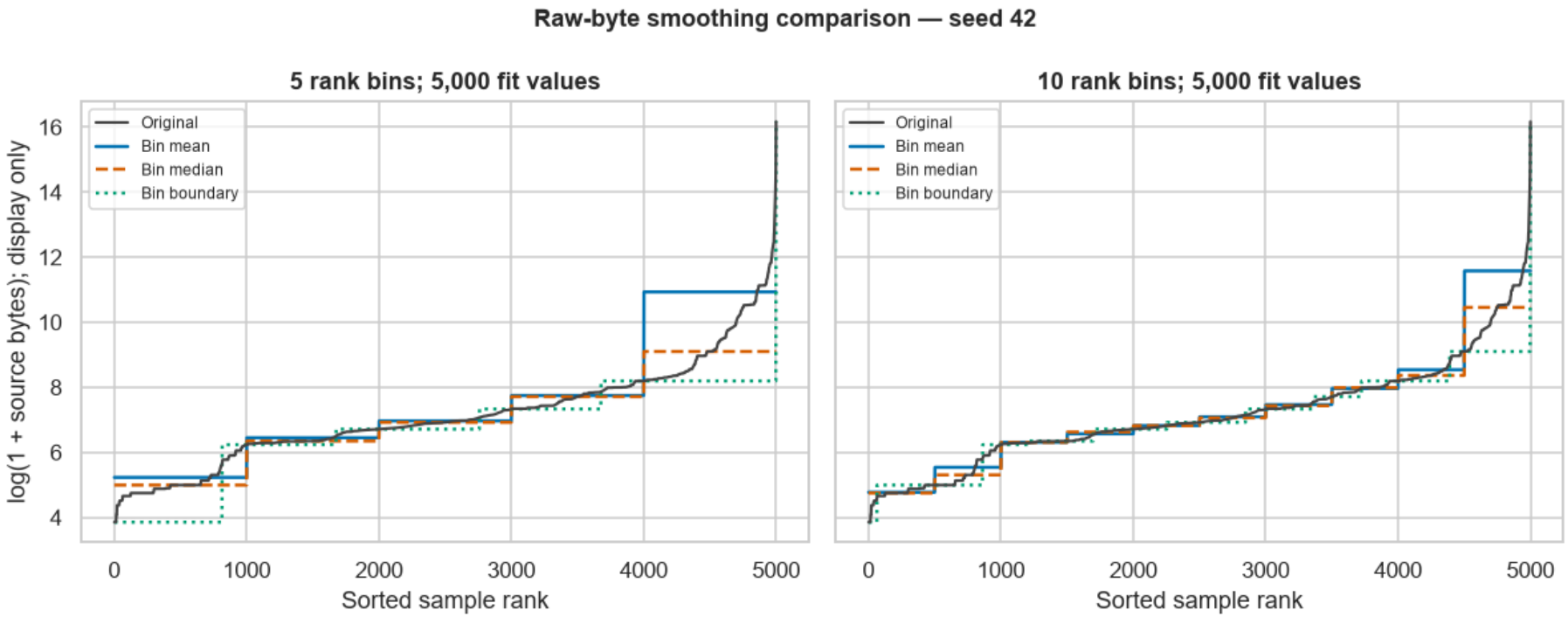


Figure 3. *A seeded sample of 5,000 fit sbytes values is smoothed into five or ten rank bins. Raw bytes are smoothed; log1p is used only to display the result. Mean smoothing is pulled upward by large transfers, while median and boundary replacements preserve different aspects of the sample.*

Mean absolute distortion (bytes)	5 bins	10 bins
mean	14930.98	13272.52
median	10319.50	9169.21
boundary	7708.97	7032.47

Stable sorting creates equal rank groups of 1,000 values (five bins) or 500 values (ten bins); equal-valued ties keep their seeded order and may cross a bin boundary. Replacements use the bin mean, median or nearest endpoint, with midpoint ties sent to the lower endpoint. Original row identity and order are restored and checked.

The last five-bin group spans 3,614 to 10,508,068 bytes: mean 55,842.54 and median 8,928. The last ten-bin group spans 8,928 to 10,508,068: mean 106,585.28 and median 34,710. Boundary smoothing keeps the largest observation at its upper endpoint. Median minimizes absolute error among single-value replacements; boundary smoothing has two candidate endpoints. These sample rank bins are not cutpoints learned for future model input.

Transactions and five frequent-pattern miners

Transactions use proto, service, state, dur, sbytes, dbytes, ct_srv_src and sttl from eligible training data. Numeric quartiles are learned from training; duplicate cutpoints are collapsed and infinite endpoints cover the range. The main seeded sample contains **5,000 transactions, eight items per transaction and 79 distinct tokens**. Each transaction is a set; all searches stop at length 3.

For example, dur=B1 means (-infinity, 0.0062005]; sbytes cutpoints are 546, 1,004 and 2,934 bytes. A sampled flow becomes: ct_srv_src=B1; dbytes=B4; dur=B2; proto=tcp; sbytes=B1; service=ftp-data; state=FIN; sttl=B1. No target is included in the primary transactions.

Miner	Implemented mechanism
Apriori	mlxtend level-wise candidate joins; all smaller subsets must be frequent; exact support counting.
FP-Growth	mlxtend compressed FP-tree and conditional pattern growth; avoids Apriori candidate generation.
Hash Apriori	Educational 4,099-bucket pair hashing prunes C2, followed by exact counts and higher-level Apriori. It does not implement full DHP transaction pruning.
DIC	Educational circular stations of 256 transactions and four states: dashed/solid, circle/box. Adds candidates between scan boundaries; every candidate receives one full lap.
Vertical TID	Educational ECLAT-style sorted-prefix recursion; intersect transaction-ID sets for extensions.

Correctness and actual mechanism evidence

The saved tests passed **575 comparisons against brute force**, including the 300-row, 12-token projected real-data check, and **15 full-feature real-data comparisons**. Tiny station sizes exercise DIC transitions; support boundaries and exact integer counts are checked. On the main .20 experiment, hashing reduced pair candidates from 190 to 32 with three full scans. DIC used 57 stations, 14,352 transaction reads (2.8704 equivalent full scans), inserted 173 candidates before the first lap ended, and finished with 66 frequent / 224 infrequent solid candidates.

Support, confidence and lift

Support(A -> B) = count(A union B)/N; confidence = count(A union B)/count(A); lift = confidence/support(B). Lift above 1 means the consequent is more common with the antecedent than overall. Minimum count is ceil(support x N). At supports .20/.30/.40, all five miners agree exactly on **66/18/13** itemsets and counts (minimum counts 1,000/1,500/2,000).

Selected rule (confidence >= .80; lift > 1.05)	Support	Confidence	Lift	Count
proto=udp -> dur=B1	0.2086	0.9304	3.562	1043
ct_srv_src=B3 -> service=-	0.2066	0.8652	1.493	1033
proto=tcp; sttl=B1 -> state=FIN	0.2888	1.0000	1.350	1444
state=FIN -> proto=tcp	0.7408	1.0000	1.320	3704

Closed means no discovered superset has the same count; maximal means no discovered frequent superset exists. At support .20, lengths 1/2/3 have 20/32/14 sets, 18/18/14 closed flags and 2/6/14 maximal flags. These flags are valid only within length <=3: unseen four-item supersets have not been ruled out.

Support/confidence experiment on the same 5,000 flows

Support	Min. count	Itemsets	Conf. .70: all / lift > 1	Conf. .80: all / lift > 1	Conf. .90: all / lift > 1
0.2	1000	66	71 / 60	55 / 55	45 / 45
0.3	1500	18	18 / 12	11 / 11	8 / 8
0.4	2000	13	12 / 9	9 / 9	6 / 6

Each rule-count cell shows all qualifying rules / those with lift above 1. Lift filters association strength relative to consequent frequency; it does not validate the rule on new traffic.

Mining experiments and measured runtime

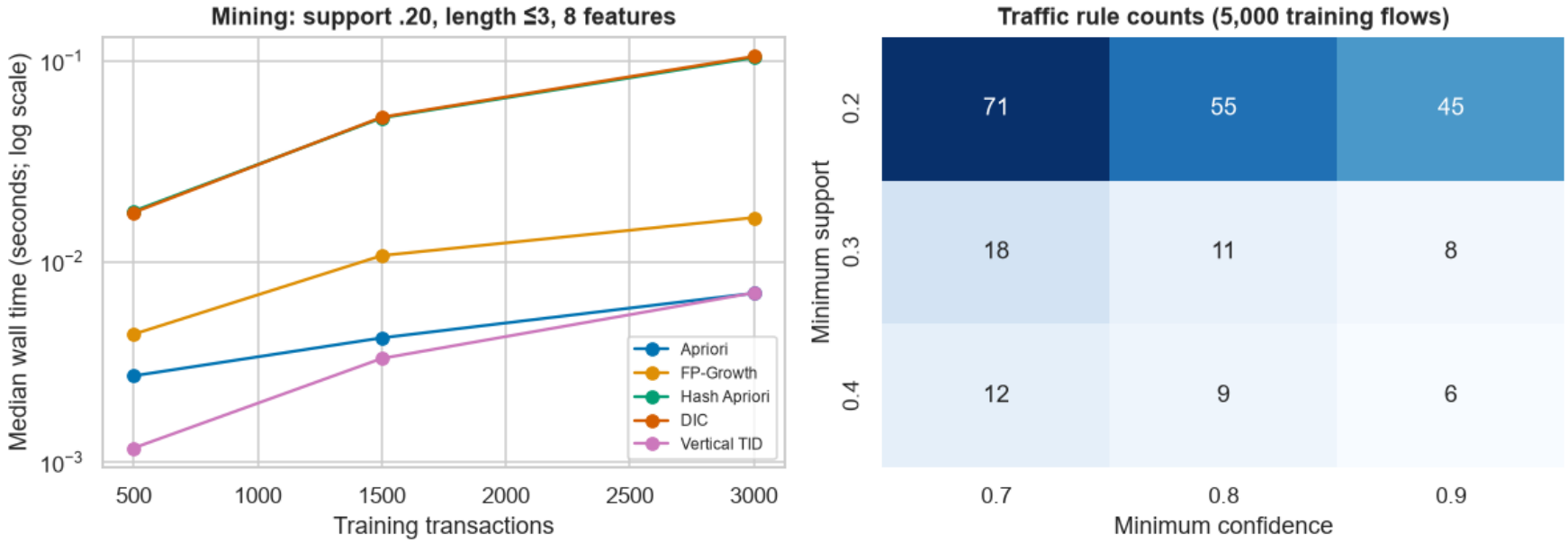


Figure 4. *The left plot compares three-repeat mining times on identical sampled transactions. The right plot counts rules for support .20/.30/.40 and confidence .70/.80/.90. Raising support removes less-common itemsets; raising confidence narrows the rules. At .20 support, rule counts fall from 71 to 55 to 45.*

Miner	500 transactions	1,500 transactions	3,000 transactions
Apriori	0.0027 (0.0027-0.0027)	0.0042 (0.0041-0.0050)	0.0070 (0.0070-0.0071)
FP-Growth	0.0044 (0.0042-0.0045)	0.0107 (0.0098-0.0118)	0.0166 (0.0165-0.0168)
Hash Apriori	0.0179 (0.0178-0.0180)	0.0520 (0.0517-0.0520)	0.1038 (0.1030-0.1055)
DIC	0.0176 (0.0162-0.0178)	0.0525 (0.0506-0.0525)	0.1053 (0.1028-0.1072)
Vertical TID	0.0012 (0.0012-0.0012)	0.0033 (0.0033-0.0036)	0.0070 (0.0069-0.0080)

Times are median (minimum-maximum) wall-clock seconds, three repetitions, support .20 and maximum length 3. Representation construction is included; file I/O, discretization, rule generation and correctness comparisons are excluded. Apriori had the smallest unrounded median at 3,000 rows; Vertical TID also rounds to 0.0070 s. Different implementation optimization and only three repeats limit broad speed claims.

A separate .05-support/.80-confidence traffic experiment produced 597 frequent sets and 412 positive-lift rules with one-item consequents. A separate training-only attack-consequent analysis found dbytes=B4; sttl=B2 -> label=attack (support .0538, confidence .9926, lift 2.0845; 269 flows). These supplementary rules describe this sample; they do not establish causality or held-out predictive performance.

Classifiers, selection and held-out results

The majority baseline predicts the most common eligible-training class (Normal). Gini and entropy trees use binary CART-style splits: Gini = 1-sum(p^2), entropy = -sum(p $\log_2$ p), with weighted impurity reduction. The entropy implementation is not ID3 or C4.5. Depth and minimum leaf size constrain overfitting.

Mixed Naive Bayes combines Gaussian numeric likelihoods and smoothed categorical likelihoods with one class prior, assuming conditional independence. Alpha smooths category counts; a learned global-scale variance floor prevents unstable Gaussian terms. Unknown-category, manual-likelihood and normalized-probability checks passed. Random Forest adds 64 bootstrap trees, square-root feature sampling and two workers (seed 42). It is the extra classification concept beyond the supplied slides.

Each tree tries depths {12,20} and leaves {5,20}; NB tries alpha {.5,1} and variance smoothing {.0001,.01}; the forest tries depths {16,None} and leaves {2,8}. Including baseline, 17 candidates are tested. Selection uses validation macro F1, then ROC-AUC, then fixed grid order. Each selected family is refitted on all eligible training rows, with no test-based retuning.

Model	Chosen parameters	Validation macro F1
Majority baseline	Most frequent	0.3402
Tree Gini	Depth 20; leaf 20	0.9192
Tree Entropy	Depth 20; leaf 5	0.9212
Mixed Naive Bayes	Alpha 0.5; variance smoothing 0.01	0.8823
Random Forest	Depth None; leaf 2	0.9319

Held-out test metrics (82,332 retained rows)

Model	Accuracy	Attack precision	Attack recall	Attack F1	Specificity	Macro F1	ROC-AUC
Majority	0.4494	0.0000	0.0000	0.0000	1.0000	0.3101	0.5000
Gini tree	0.8737	0.8249	0.9782	0.8951	0.7457	0.8683	0.9584
Entropy tree	0.8778	0.8379	0.9647	0.8968	0.7713	0.8735	0.9357
Mixed NB	0.7793	0.7173	0.9889	0.8315	0.5225	0.7559	0.8652
Forest	0.8712	0.8170	0.9873	0.8941	0.7290	0.8649	0.9814

Attack is the positive class. Precision=TP/(TP+FP), recall=TP/(TP+FN), specificity=TN/(TN+FP), F1=2PR/(P+R), and accuracy=(TP+TN)/N. Macro F1 averages the two class F1 values; undefined scores are zero. ROC-AUC uses attack scores over thresholds. No probability calibration or operational cost threshold is fitted.

Model	Fit seconds	Predict seconds	Probability seconds
Majority	0.0024	0.0008	0.0005
Gini tree	6.8458	0.2248	0.2192
Entropy tree	7.0935	0.2078	0.2000
Mixed NB	0.7935	0.1789	0.1983
Forest	54.6030	0.7441	0.6923

Measured host: Windows 11, CPython 3.13.12, 12 logical CPUs, 15.2 GiB RAM. Times include each complete model pipeline call and depend on this host and its load. Forest fitting took 54.6030 s and label prediction 0.7441 s; it took 8.0 times the Gini tree fitting time.

Confusion matrices and error tradeoffs

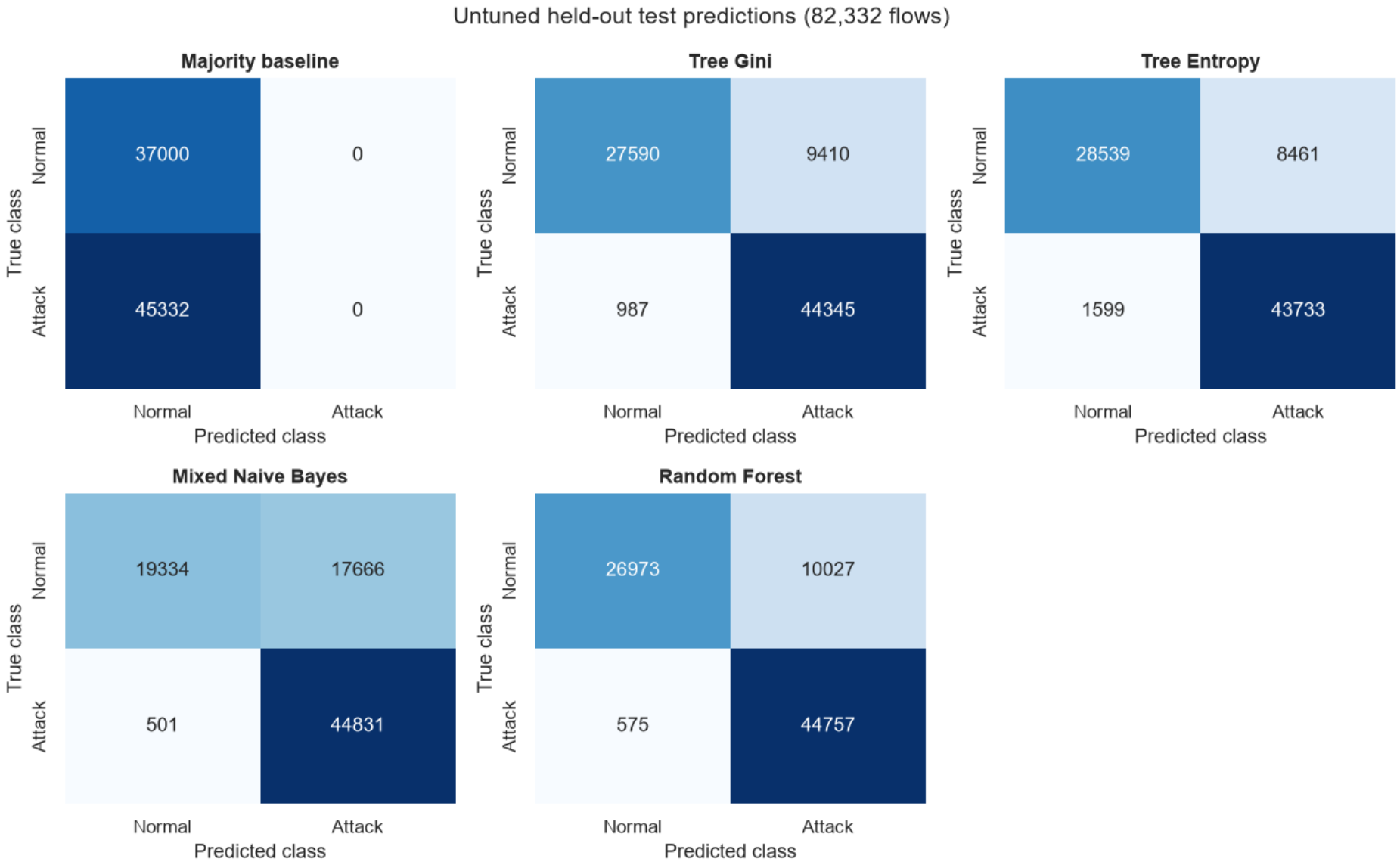


Figure 5. *Rows are true class and columns are predicted class (Normal, Attack). The selected forest has 26,973 true negatives, 10,027 false positives, 575 false negatives and 44,757 true positives. Mixed NB misses only 501 attacks but flags 17,666 normal flows. The Normal-only majority baseline misses all 45,332 attacks.*

ROC curves, model choice and feature importance

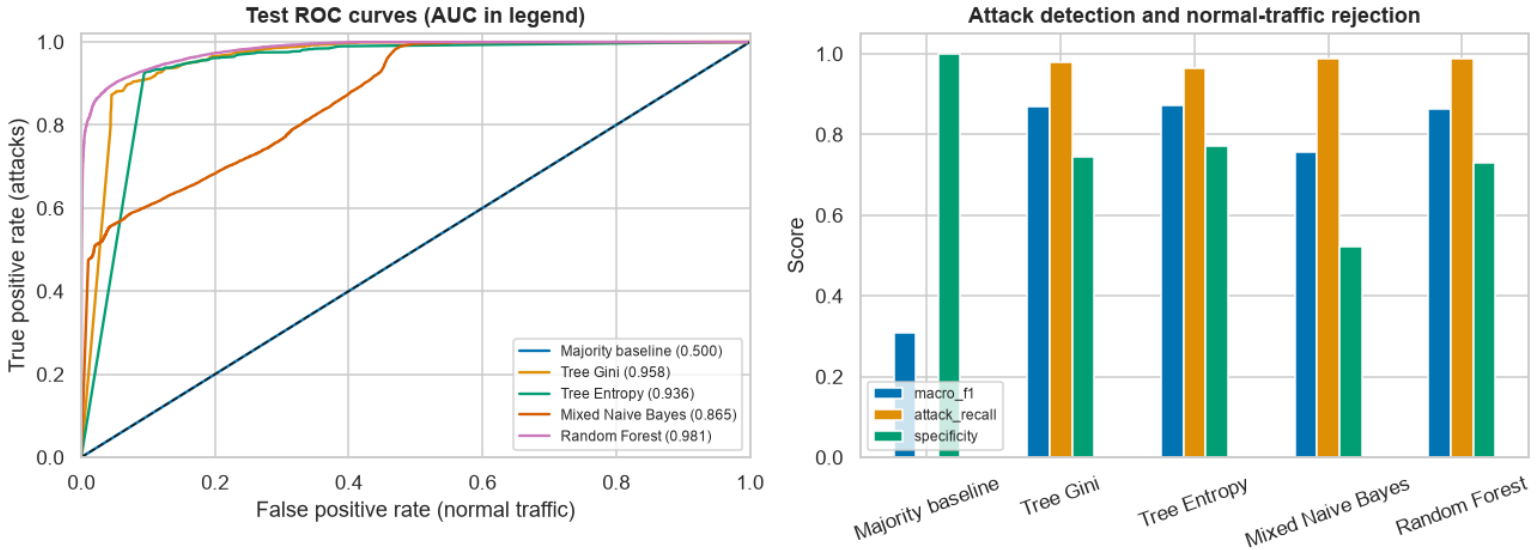


Figure 6. *The forest has the highest test ROC-AUC (0.9814), while the entropy tree has the highest test accuracy (0.8778) and macro F1 (0.8735). The forest remains the validation-selected model; test differences are reported without changing that choice.*

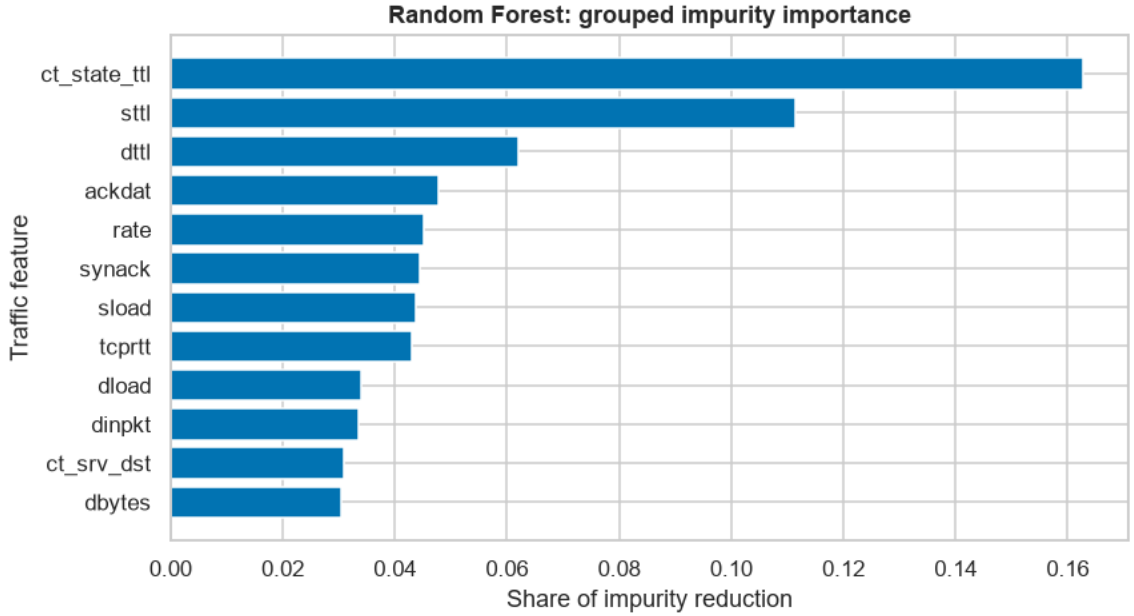


Figure 7. Grouped forest importance

The final forest uses 40 raw features, 63 encoded columns and 64 trees. One-hot category importances are summed back into their raw feature.

ct_state_ttl (0.1628), sttl (0.1115) and dttl (0.0620) rank highest. Timing, rate and load attributes also contribute.

Impurity importance may favour attributes with many possible values. Correlation can distribute importance among related features; these values describe this model and do not establish causes.

Validation forest macro F1 was 0.9319 (AUC 0.9852). On test, the entropy tree leads macro F1 by 0.0086, while NB has the highest attack recall (0.9889). A model can rank flows well yet produce more false alarms at its default decision.

Course coverage, findings and conclusion

Course concept	Slide PDF pages	Evidence / scope
KDD / statistics; quality; feature reduction	19-61	Implemented: EDA, cleaning, identities and fit-only selection.
Transformations; normalization	62-64	Primary transforms used; min-max/decimal comparison demonstrated only.
Bin smoothing; IQR / skewness	57, 64; 30-44	Demonstration / diagnostics only; no primary-input changes.
Apriori / FP-Growth	70-90	Library implementations, exact supports and rule grid.
DIC; hash Apriori; vertical TID	91-110	Educational implementations; hash variant is not full DHP.
Closed / maximal frequent sets	114-124	Implemented post-analysis only within length <=3.
Gini / entropy trees; evaluation	133-149	Implemented CART trees, metrics, confusion matrices and ROC.
Naive Bayes / smoothing	150-158	Mixed NB implemented and checked.
Random Forest	Additional	Implemented extra classification concept beyond supplied slides.
ID3, C4.5 gain ratio, SLIQ	136, 140	Explained / identified only.
Full DHP; partition/sampling variants; A-Close; Pincer	98, 111-132	Not implemented; no equivalence claim.

Findings, limitations and conclusion

Five different mining mechanisms agree exactly under common support and length limits. Higher support/confidence shrinks the discovered patterns. Classification improves strongly over the baseline, but high attack recall comes with false alarms. On 53,946 unique test identities, forest accuracy falls to 0.8229 and macro F1 to 0.8208 without retuning: retained test repetitions affect metric weights.

The data is a controlled cyber-range partition, not live institute traffic. Binary scores do not measure each attack family. A single validation split, small grid and uncalibrated probabilities limit inference; flow aggregates do not establish real-time operation. NB assumptions are approximate. Mining uses eight attributes, 5,000 rows and length <=3, so rare or longer patterns can be missed. Local CPU timings do not establish distributed scalability.

The project connects the course KDD process to an executed, checked experiment. Its main conclusion is that data population, operating decisions and evaluation criteria matter alongside the algorithm name: the validation choice, test metric leaders and computational costs tell different parts of the result.

Sources and compact dataset bibliography

[1] UNSW-NB15 official documentation. [2] Mireu-Lab pinned mirror. [3] Iacg030175 standard configuration.
[4] Supplied course slides: BigData_mid+end_compressed.pdf (224 pages); page numbers above are PDF positions. Implementation/results: Network_Traffic_Project.ipynb, saved execution 10 October 2026.

[5] Essential implementation documentation: scikit-learn trees, Naive Bayes, forests; mlxtend Apriori / FP-Growth. Mixed NB and the three educational miners are defined in the notebook.

Dataset publications requested by the official source (compact attribution):
1. Moustafa, N.; Slay, J. (2015). UNSW-NB15: a comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set). MilCIS.
2. Moustafa, N.; Slay, J. (2016). The evaluation of Network Anomaly Detection Systems: Statistical analysis of the UNSW-NB15 data set and the comparison with the KDD99 data set. Information Security Journal.
3. Moustafa, N. et al. (2017). Novel geometric area analysis technique for anomaly detection using trapezoidal area estimation on large-scale networks. IEEE Transactions on Big Data.
4. Moustafa, N. et al. (2017). Big data analytics for intrusion detection system: statistical decision-making using finite Dirichlet mixture models. Data Analytics and Decision Support for Cybersecurity, pp. 127-156.
5. Sarhan, M.; Layeghy, S.; Moustafa, N.; Portmann, M. (2020). NetFlow Datasets for Machine Learning-Based Network Intrusion Detection Systems.